Renjie LIU

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EDUCATION

Southern University of Science and Technology **Sep. 2023 – Jun. 2026**
*M.phil. in Computer Science — Advised by **Prof. Bo Tang*** *ShenZhen, China*

- GPA: 3.62/4.00 (90/100)

Southern University of Science and Technology **Sep. 2019 – Jun. 2023**
*B.Eng. in Computer Science — Advised by **Prof. Bo Tang** & **Prof. Xiao Yan*** *ShenZhen, China*

- GPA: 3.82/4.00 (91/100)
- Key Coursework: Linear Algebra (93), Discrete Math (96), Algorithm Design & Analysis (97), Database Principles (91), Computer Organization (94), Computer Network (96), Artificial Intelligence (92)
- Language Proficiency: TOEFL (104), CET-6 (590)

EXPERIENCE

NYU System Group, New York University **Dec. 2024 – Jun. 2025**
*Student Intern — Advised by **Prof. Jinyang Li***

- General and effective **GraphRAG** solutions and robust **Agent** program development.

Shanghai AI Lab, Amazon Web Services **Jun. 2022 – Nov. 2024**
*Applied Scientist Intern in **DGL Team** — Advised by **Dr. Minjie Wang***

- Efficient **GNN training** on large-scale datasets in single-machine and distributed environments.

PUBLICATIONS

- **PolyG: Effective and Efficient GraphRAG with Adaptive Graph Traversal**
  Renjie Liu, Haitian Jiang, Xiao Yan, Bo Tang, and Jinyang Li
  https://arxiv.org/pdf/2504.02112
- **DiskGNN: Bridging I/O Efficiency and Model Accuracy for Out-of-Core GNN Training**
  Renjie Liu*, Yichuan Wang*, Xiao Yan, Zhenkun Cai, Minjie Wang, Haitian Jiang, Bo Tang, and Jinyang Li
  *ACM Conference on Management of Data **(SIGMOD 2025)***
  *Equal Contribution
- **APT: Adaptive Parallel Training for Graph Neural Networks**
  Kaihao Ma*, **Renjie Liu***, Zhenkun Cai, Xiao Yan, Xiang Song, Minjie Wang, and James Cheng
  *ACM Symposium on Principles and Practice of Parallel Programming **(PPoPP 2025)***
  *Equal Contribution
- **Forming Scalable, Convergent GNN Layers that Minimize a Sampling-Based Energy**
  Haitian Jiang, **Renjie Liu**, Xiao Yan, Zengfeng Huang, Zhenkun Cai, Minjie Wang, and David Wipf
  *International Conference on Learning Representations **(ICLR 2025)***
- **gSampler: General and Efficient GPU-based Graph Sampling for Graph Learning**
  Ping Gong, **Renjie Liu**, Zunyao Mao, Zhenkun Cai, Xiao Yan, Cheng Li, Minjie Wang, and Zhuozhao Li
  *ACM Symposium on Operating Systems Principles **(SOSP 2023)***

TECHNICAL SKILLS

Programming Languages: Python (system abstraction), C/C++ (core operators), CUDA (GPU kernels).
Frameworks and Tools: vLLM, Pytorch, DGL, PyG, Thrust, SQL, Latex, Git, etc.

RESEARCH PROJECTS

Efficient KV Cache Reuse for Serving LLMs with Mixed Compression — **Aug. 2025 – Present**

- Exploring a new KV cache compression mechanism for fast LLM serving while preserving the same model accuracy in RAG scenarios, where chunk-caches are reused across queries with selective recomputation.
  - ✏ Observe that chunk-caches of knowledge base can be large, and existing systems are dominated by disk I/O transfers.
  - ✏ Exploring a mixed-precision approach to compress KV caches without model performance degradation.

Towards Reliable and Transparent Web AI Agent — **May 2025 – Present**

- Exploring a new execution paradigm for web agent by modularizing general REACT agents into co-operative components and regulating agent's action by explicit query planning with action templates.
  - ✏ Observe that existing agents generally follow the REACT paradigm, which is prone to failure and difficult to debug.
  - ✏ Exploring a new abstraction to modularize the agent's execution with improved reliability and transparency.

Effective and Efficient Graph RAG Solution (PolyG) — **Aug. 2024 – May 2025**

- Construct a new benchmark tailored for Graph RAG queries and propose a new approach that is both effective and efficient with adaptive graph traversal under diverse query patterns.
  - ✏ Observe that existing GraphQA benchmarks lack query diversity, and create the first Graph RAG benchmark.
  - ✏ Propose a Graph RAG solution that adaptively selects the best traversal methods based on the query pattern.

Efficient out-of-core GNN Training on Large-scale Graphs (DiskGNN) — **Sep. 2023 – July 2024**

- Build a system which achieves high I/O efficiency and thus fast training without hurting model accuracy for out-of-core GNN training where the whole graph can not fit in CPU memory.
  - ✏ Observing that offline sampling scheme does not degrade model accuracy by sampling mini-batches beforehand.
  - ✏ Propose a suite of designs based on offline sampling, including four-level feature store and batched feature packing.

Adaptive Parallel Training for Graph Neural Networks (APT) — **April 2023 – Dec. 2023**

- Build a general framework capable of automatically determine the best parallelism for distributed GNN training based on hardware environments and models.
  - ✏ Derive an abstraction for all parallel schemes from the perspective of matrix segmentation & multiplication.
  - ✏ Design a cost model to automatically determine the best parallelism given specific environments and configurations.

General and Efficient GPU-based Graph Sampling (gSampler) — **Aug. 2022 – April 2023**

- Build a high-performance GPU sampling framework for GNN training that can express diverse sampling algorithms by sketching sampling process using matrix operations.
  - ✏ Propose matrix-centric APIs based on a proposed 4-step ECSF model to sketch the graph sampling process.
  - ✏ Incorporate a suite of sampling-oriented optimizations, e.g., kernel fusion, format selection and batching.

HONORS AND AWARDS

- China National Scholarship — **Nov. 2024**
- University Merit Scholarship for Outstanding Student: The 1st Class — **Sep. 2023, 2022**
- Outstanding Graduates of Department of Computer Science and Engineering — **Jun. 2023**
- ACM SIGMOD 2022 Programming Contest: World's Finalist — **April 2022**

TEACHING EXPERIENCE

Teaching Assistant

- **Computer Organization and Architecture** — **Mar. 2022 – Jun. 2022**
  - ❍ Designed and graded course assignments and the course project (a Minisys CPU written in Verilog on FPGA).
  - ❍ Answered questions about CPU architecture, MIPS and Verilog programming in laboratory courses.
- **Introduction to Computer Programming** — **Sep. 2020 – Jun. 2021**
  - ❍ Designed and graded course assignments and the course project (a Minesweeper implemented by Java).
  - ❍ Answered questions about programming skills and algorithms in laboratory classes.